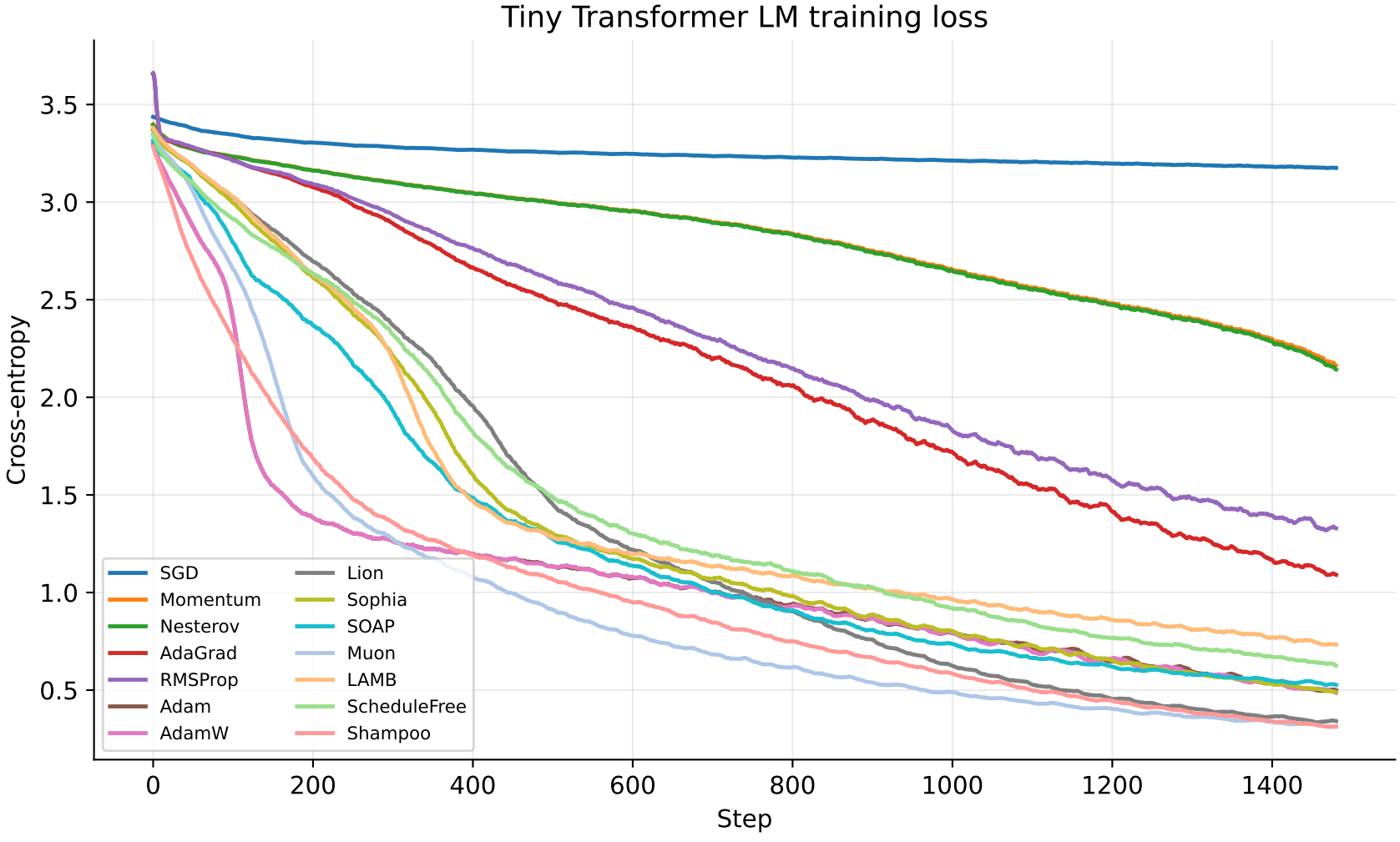

Tiny Transformer LM training loss
Cross-entropy
Step
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0
200
400
600
800
1000
1200
1400
SGD
Momentum
Nesterov
AdaGrad
RMSProp
Adam
AdamW
Lion
Sophia
SOAP
Muon
LAMB
ScheduleFree
Shampoo